WEEK-4

Question 1:

Solve the following problem with CPU scheduling algorithms.

Draw a table with 4 rows & 3 columns having the following details.

Process	AT	BT
P_1	0	5
P_2	0	4
P_3	0	3

There is a Context Switch in the CPU the Context Switch time is 5 seconds. Consider the Context switch time b/w processes of five seconds.

Apply the following Algorithms:-

i) preemptive algorithms:-

(i) SRTF (ii) Priority scheduling

2) Non-Preemptive ~~Sche~~ Algorithms-

(i) FCFS

(ii) SJF

(iii) Priority scheduling

Solve each algorithm with its Grand chart, CT, TAT, WT, average TAT and average WT

Given,

Process	AT	BT
P_1	0	5
P_2	0	4
P_3	0	3

→ Context switch time = 5sec

TAT = CT − AT

WT = TAT − BT

(i) FCFS

order:- $P_1 \rightarrow P_2 \rightarrow P_3$

Grantt Chart

0	P_1	5	CS	10	P_2	14	CS	19	P_3	22

Process	AT	BT	CT	TAT=CT-AT	WT=TAT-BT
P_1	0	5	5	5	0
P_2	0	4	14	14	10
P_3	0	3	22	22	19

Average WT= $\frac{0+10+19}{3} = 9.67$

Average TAT = $\frac{5+14+22}{3} = 13.67$

(iii) SJF

$P_3(3) \rightarrow P_2(4) \rightarrow P_1(5)$

Order: $P_3 \rightarrow P_2 \rightarrow P_1$

Gantt Chart:

0	P_3	3	CS	8	P_2	12	C_2	17	P_1	22

Process	AT	BT	CT	TAT	WT
P_1	0	5	22	22	17
P_2	0	4	12	12	8
P_3	0	3	3	3	0

$$\text{Average WT} = \frac{17+8+0}{3} = 8.33$$

$$\text{Average TAT} = \frac{22+12+3}{3} = 12.33$$

(iii) SRTF

Order: $P_3 \rightarrow P_2 \rightarrow P_1$

Gantt chart:

0	3	8	12	17	22
P_3	CS	P_2	CS	P_1	

Process	AT	BT	CT	TAT	WT
P_1	0	5	22	22	17
P_2	0	4	12	12	8
P_3	0	3	3	3	0

Average TAT = 12.33

iv) Priority scheduling:

Cannot solve because priority values are not given.

Question 8:

A mobile operating System has the following tasks

Task	AT	CPU Burst
Music Player (MP)	0	20
Touch Event (TE)	3	1
whatsapp Notification (WN)	5	2
Camera Launch (CL)	6	4

Questions:

(i) which scheduling algorithm is most suitable FCFS, SJF, SRTF (or) RR?

(ii) Draw the execution timeline for SRTF

(iii) Explain why the selected algorithm improves user experience

(iv) which task receives the fastest response time

(i) SRTF is the suitable scheduling algorithm because SRTF always selects the task with shortest remaing CPU burst time

(ii) SRTF Execution Timeline:

In SRTF, the process with the shortest remaining burst time is selected.

0-3: MP runs

Remaining MP = 20-3 = 17

At 3: TE arrives with BT = 1

TE is shorter than MP → TE runs from 3-4

4-5: MP continues

Remaining MP = 16

At 5: WN arrives with BT = 2

WN is shorter than MP → WN runs

5-7:- WN completes

7-11: Compare MP = 16 & CL = 4 → CL runs

11-27: CL finishes, so remaining MP runs unit completion.

Grant Chart:

0 | MP | 3 | TE | 4 | MP | 5 | WN | 7 | CL | 4 | MP | 27

→ Final Execution Order

MP → TE → MP → WN → CL → MP

(iii) SRTF gives quick response to short tasks such as touch events, notifications and Camera launch.

Therefore, the mobile device feels:-

* Fast
* ~~Respo~~ Responsive
* Interactive

(iv) Response Time = First CPU Start time − AT

TASK	AT	BT	first CPU start	CT	TAT	WT	RT
MP	0	20	0	27	27	7	0
TE	3	1	3	4	1	0	0
WN	5	2	5	7	2	0	0
CL	6	4	7	11	5	1	1

where,

TAT = CT − AT ; WT = TAT − BT

question 3:

Define the following terms:-

(i) starvation

Starvation occurs when a process waits for a very long time because other processes are repeatedly selected for execution.

(ii) Convoy effect

Convoy effect occurs in FCFS Scheduling when a long process makes many short processes wait behind it

(iii) Context Switching.

It is the process of saving the state of the currently running process and loading the state of another process so that the CPU can switch between them.

(iv) Aging

Aging is a technique used to prevent starvation by gradually increasing the priority of the process that has been waiting for a long time.

Question 4:

Consider the following process.

Process	AT	BT	Priority
P_1	0	4	3
P_2	0	9	1
P_3	0	6	2
P_4	0	4	3
P_5	0	9	1
P_6	0	6	2
P_7	1	4	3
P_8	1	9	1
P_9	2	6	2

(i) FCFS

FCFS means first come first serve

0	4	13	19	23	32	38	42	51	55
P_1	P_2	P_3	P_4	P_5	P_6	P_7	P_8	P_9	

Process	AT	BT	CT	TAT	WT	RT
P_1	0	4	4	4	0	0
P_2	0	9	13	13	4	4
P_3	0	6	19	19	13	13
P_4	0	4	23	23	19	19
P_5	0	9	32	32	23	23
P_6	0	6	38	38	32	32
P_7	1	4	42	41	37	37
P_8	1	9	51	50	41	41
P_9	2	6	57	55	49	49

Average TAT = 30.56.

Average WT = 24.22

(ii) SJF

SJF means Shortest Job first. Since it is non-preemptive, once a process starts, it finishes completely.

Order:

Shortest available processes are selected

$P_1 \rightarrow P_4 \rightarrow P_7 \rightarrow P_3 \rightarrow P_6 \rightarrow P_9 \rightarrow P_2 \rightarrow P_5 \rightarrow$

Grantt chart

0	4	8	12	18	24	30	39	48	57
P_1	P_4	P_7	P_3	P_6	P_9	P_2	P_5	P_8	

Process	AT	BT	CT	TAT	WT	RT
P_1	0	4	4	4	0	0
P_4	0	4	8	8	4	4
P_7	1	4	12	11	7	7
P_3	0	6	18	18	12	12
P_6	0	6	24	24	18	18
P_9	2	6	30	28	22	22
P_2	2	9	39	39	30	30
P_5	0	9	48	48	39	39
P_8	1	9	56	56	47	47

Average TA

Average WT = 19.89

(iii) SRTF

SRTF means Shortest Remaining Time first

→ It is the premptive Version of SJF

Process	CT	TAT	WT	RT
P_1	4	4	0	0
P_4	8	8	4	4
P_7	12	11	7	7
P_3	18	18	12	12
P_6	24	24	18	18
P_9	30	28	22	22
P_2	39	39	30	30
P_5	48	48	39	39
P_8	57	56	47	47

Average TAT = 26.22

Avg WT = 19.89